



picoEmerald FT

Synchronously Pumped OPO

User Manual



Manual of picoEmerald FT

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IMPORTANT - READ CAREFULLY BEFORE USE - KEEP FOR FUTURE REFERENCE

This user manual contains user information for the picoEmerald FT. Read this manual carefully before operating the picoEmerald FT. The picoEmerald FT has only to be used as described in this manual. Differing use may endanger safety and voids warranty.

CAUTION - USE OF CONTROLS OR ADJUSTMENTS OR PERFORMANCE OF PROCEDURES OTHER THAN THOSE SPECIFIED HEREIN MAY RESULT IN HAZARDOUS RADIATION EXPOSURE

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Symbols Used in This Manual



This symbol is intended to alert the operator to the danger of exposure to hazardous visible and invisible laser radiation.



This symbol is intended to emphasize important operating instructions.



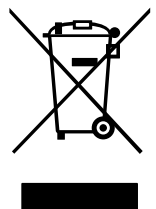
This symbol is intended to inform the operator about important or useful but not hazard-related information.

Warranty

The warranty conditions are specified in the sales contract. Any unauthorized modification of the picoEmerald FT system components or software will void the warranty and service contract.

Disposal Hints

All electrical and electronic products should be disposed separately from the standard municipal waste system. Proper disposal of your old appliance prevents potential negative consequences for the environment and human health.



Some components of your picoEmerald FT system are marked with the crossed-out wheeled bin symbol covered by the European Directive 2002/96/EC on waste electrical and electronic equipment (WEEE) of the European Parliament and the Council of January 27, 2003. These items must be disposed via designated collection facilities appointed by government or local authorities.

Declaration of Conformity to EU RoHS

Products manufactured by APE Angewandte Physik und Elektronik GmbH are in compliance with EU Directive 2015/863 of the European Parliament and of the Council of July 22, 2019 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (also known as **RoHS Recast**). In addition, this declaration of conformity is issued under the sole responsibility of APE Angewandte Physik und Elektronik GmbH. Specifically, products manufactured do not contain the substances listed in the table below in concentrations greater than the listed maximum value.

Substance	Limit (ppm)
cadmium (Cd)	< 100
lead (Pb)	< 1000
mercury (Hg)	< 1000
hexavalent chromium (Cr VI)	< 1000
polybrominated biphenyls (PBB)	< 1000
polybrominated diphenyl ethers (PBDE)	< 1000
bis(2-Ethylhexyl) phthalate (DEHP)	< 1000
benzyl butyl phthalate (BBP)	< 1000
dibutyl phthalate (DBP)	< 1000
diisobutyl phthalate (DIBP)	< 1000

Signature:

A handwritten signature in blue ink, appearing to read 'K. v. Volkmann', is shown on a light blue background.

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Title: CEO

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1 Safety Instructions

1.1 Safety Features and Compliance to Government Requirements

US government requirements for this product are contained in CFR, Title 21, Chapter 1, Subchapter J, administered by the Center for Devices and Radiological Health (CDRH).

The European Community requirements for this product safety are specified in the “Low Voltage Directive” (2014/35/EU) and in harmonized standards. Compliance of this product is certified by the CE mark.

1.2 Electrical Safety

The picoEmerald FT uses DC (OPO unit) and AC (laser controller) voltages. All units are designed to be operated with protective covers in place. The device complies with protection Class I / EN 61140:2007, degree of protection IP20, according to EN 60529:2019-06.



To run the picoEmerald FT, the provided power supply must be used.



Be aware that the laser uses AC voltage. To run the laser, the provided power supply must be used.



Users are not recommended to open the picoEmerald FT housing. Opening the housing is only allowed for trained service personnel. If it is necessary to open the housing for service purposes (by trained service personnel!), the device has to be disconnected from the power supply.

1.3 Electromagnetic Compatibility

In the EMC Directive 2014/30/EU, the European requirements for Electromagnetic Compliance (EMC) are specified. Conformance (EMC) is achieved through compliance with the harmonized standards EN 61326-1:2013.

With respect to the disturbance voltage or field strength, the picoEmerald FT meets the requirements for Class B as specified in EN 55011:2011-04.

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1.4 Cooling Fluids

Cooling fluid is supplied with the picoEmerald FT. Please be aware that this cooling fluid is harmful if swallowed or inhaled. Also avoid the liquid coming into contact with skin or eyes. The warning information of the cooling fluid's merchant is supplied in addition to this manual. Use extreme caution when handling the cooling fluid and keep the merchant's warning in mind.

1.5 Optical Safety

Because of its special properties, laser light poses safety hazards not associated with light from conventional sources. The safe use of lasers requires that all laser users - and everyone else near the laser system - are aware of the dangers involved. The safe use of the laser depends upon the user becoming familiar with the instrument and the properties of intense and coherent beams of light.



Exposing the eye directly to the output beam of a laser will cause serious damage and possible blindness.

In addition to the main beam, there are often many smaller beams present at various angles near the laser system. These beams are formed by specular reflections of the main beam at polished surfaces, e.g., such as lenses and beam splitters.



Although weaker than the main beam, such beams may still be sufficiently intense to cause eye damage.

Laser beams are powerful enough to burn skin, clothing, or paint. They can ignite volatile substances such as alcohol, gasoline, ether, and other solvents, and can damage light-sensitive elements in video cameras, photomultipliers, and photodiodes.

The laser beam can ignite substances in its path, even at a distance, for example if these are highly flammable. Besides the damage and injury by the direct contact with the laser beam, both can also be caused if contacted indirectly from reflective surfaces. For these and other reasons, the user is advised to follow the precautions below:

1. Observe all safety precautions in the user manual.
2. Extreme caution should be exercised when using flammable material in the area of the laser. These must not cross or even be placed in the laser beam's path.
3. Limit access to the laser to qualified users who are familiar with laser safety practices and who are aware of the dangers involved.
4. Never look directly into the laser light source or at scattered laser light from any reflective surface, even not if wearing appropriate laser safety glasses.
5. Maintain experimental setups parallel to the optical table at low heights to prevent inadvertent beam-eye contact at eye level. Also, do not bend down while the laser system is running.
6. The laser beam should be enclosed by laser beam shielding covers so that the section where the laser beam is free accessible is as small as possible.

7. As a precaution against accidental exposure to the output beam or its reflection, those using the system must wear safety glasses that are appropriate for the specifications of the laser being used.



Laser safety glasses protect the human eyes from potentially damaging exposure. Even if laser safety glasses are used, the operator must not expose the eyes directly to the laser beam. In addition, the operator should use extreme caution even when using safety glasses because the light at the laser wavelengths are blocked which prevents the operator from seeing the beam.

8. Avoid direct or indirect exposure to the laser light. The intensity of the beam can easily cause skin burns or ignite clothing.
9. Use the laser in an enclosed room with impenetrable walls and windows for the whole optical spectrum. Laser light will remain collimated over long distances and therefore presents a potential hazard if not confined.
10. To alert those who are present, post warning signs in the whole area where direct and indirect laser light can occur.
11. Advise all those using the laser of these precautions. It is good practice to operate the laser in a room with controlled and restricted access.

1.6 Laser Classification

The governmental standards and requirements specify that the laser must be classified according to the output power or energy, and the laser wavelength. The picoEmerald FT is classified to emit laser radiation Class 4 based on CFR, Title 21, Chapter 1, Subchapter J, Section 1040.10(d).

According to the European Community standards, the picoEmerald FT is classified to emit laser radiation Class 4 based on EN 60825-1:2022-07, Clause 5. In the manual and other documentation of the picoEmerald FT, the classification will be referred to as Class 4.

The pump laser contained in this system meets the following specifications.

wavelength	1032 nm
beam divergence	1.5 mrad $\pm$ 0.2 mrad
pulse duration	< 2.3 ps
repetition rate	40 MHz
maximum power or energy output	20 W

1.7 Protective Housing

The picoEmerald FT is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class 1 radiation, as specified in CFR, Title 21, Chapter 1, Subchapter J, Section 1040.10(f)(1) and EN 60825-1:2022-07, Clause 6.2, except for the output beam, which is laser radiation Class 4 dependent upon the pump or input laser.

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Opening the housing requires tools and the operator must not open the housing at any time!

1.8 External Interlock Connector and Manual Reset Mechanism

The picoEmerald FT is equipped with an external interlock connector. In case the interlock is open, the system shutter will close automatically and laser emission is terminated. Laser emission does not resume until the laser system is manually reset as described in section . [Complies to CFR, Title 21, Chapter 1, Subchapter J, Section 1040.10(f)(3) & (10) and EN 60825-1:2022-07, Clause 6.4 & 6.5].



The picoEmerald FT is a Class 4 laser. Due to safety regulations it must be connected to a door switch interlock to prevent accidents. The door switch interlock enhances safety, as it shuts off laser emission if the door to the room or enclosure where the system is located is opened.

Notice: Do not short-circuit the interlock input. This will allow laser emission from the picoEmerald without the door switch interlock, which is against safety regulations. APE does not take liability for any issues due to bypassing the interlock safety mechanism.

Interlock Connector: A cable connector for the interlock interfacing on the back of the system is provided with the picoEmerald laser. This plug is a LEMO connector type FGG.0B.302.



Figure 1.1 Interlock plug

Interlock Power Fail: Please ensure that there is no short connection from the interlock signals to ground/chassis as this will generate an Interlock power fail, which will disable the system. If there is an Interlock power fail, then switch off the mains supply, locate the short connection from the interlock signal to ground and remove it. Turn on the picoEmerald system and reset interlock.

Cable: The Interlock cable can be up to 5 meters long and it can be a non-shielded type. If a cable longer than 5 meters is required we recommend using a shielded type of cable.

1.9 Key Control

Without the laser key no laser emission is possible. The key cannot be pulled out in ON (Armed) position. [Complies to CFR, Title 21, Chapter 1, Subchapter J, Section 1040.10(f)(4) and EN 60825-1:2022-07, Clause 6.6].

1.10 Laser Radiation Emission Indicators

When the internal laser source of the device is activated, the laser emission indicator on the front of the device illuminates white [Complies to CFR, Title 21, Chapter 1, Subchapter J, Section 1040.10(f)(5)/EN 60825-1:2022-07, Clause 6.7]. The position is indicated with (3) at figure 1.2.

1.11 Location of Information and Safety Labels

There exists laser and collateral radiation inside the picoEmerald FT. The picoEmerald FT is designed to be used with the covers in position and these covers shield the operator from all collateral radiation.



It is only allowed to remove the covers by service personnel for service operation.



Figure 1.2 Safety label on the picoEmerald FT. The numbers in the figure correspond to the numbers in the following list

1.

LASER APERTURE

Through this aperture / these apertures laser radiation is emitted.

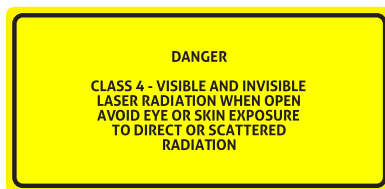
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2.



Laser safety label showing the laser safety class and the maximum amount of light emitted

3.



Warning label indicating that dangerous light is emitted from the device when the cover is removed for servicing.

2 Description

2.1 Description and Intended Use

The picoEmerald FT is a laser light source specifically made for SRS (Stimulated Raman Scattering) and CARS (Coherent Anti-Stokes Raman Spectroscopy) microscopy. An integrated fiber laser delivers a beam of picosecond laser pulses at 1032 nm that is split internally. One part is frequency doubled to synchronously pump an internal OPO (Optical Parametric Oscillator) and the other part is used as the SRS-Stokes beam. The Signal - also named SRS-Pump - output beam of the picoEmerald FT is generated by the OPO. It is designed to provide ultrashort picosecond laser pulses in the wavelength range between 660 ... 1010 nm. Optionally, an OPO Idler beam that covers the range between 1055 ... 2340 nm is available in addition.



Figure 2.1 picoEmerald FT optical unit

The picoEmerald FT supplies full automation to overlap the two ultrafast pulse trains in time and space: the 1032 nm out of the laser oscillator and the wavelength tunable OPO Signal. One common output port is used for the selected pulse train or the combined trains while the optional Idler beam exits through a separat port.

Independent power adjustments of the 1032 nm beam and the OPO Signal, sensors for the spatial and temporal overlap, and a high resolution spectrometer are integrated. These features make it an ideal laser source for experiments that require synchronous pulses at adjustable different wavelengths such as CARS and coherent Raman microscopy.

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Optionally an amplitude modulation of the 1032 nm beam with 20 MHz through an electro-optical modulator (EOM) is available for use with SRS or heterodyne CARS microscopy to suppress the non-resonant background.

In addition there are auxiliary laser output ports for the fundamental laser beam (1032 nm, vertical polarized) and for the frequency doubled beam (516 nm, horizontal polarized) at the back side of the picoEmerald FT (See figure 2.4). These extra output beams are optional.

The picoEmerald FT is intended to be operated under controlled laboratory condition, that is in closed, dry, low-dust and temperature stabilized rooms after installation on an optical table or a similar stable vibration-free base.

The picoEmerald FT is delivered with an internal closed-loop water chiller unit, to maintain stable operation. Its optical unit is designed for 'hands-free' operation, so it isn't necessary for the user to have any interaction with the OPO other than via the software interface.

2.2 Specifications

2.2.1 Optical Parameters

	OPO-Signal (SRS-Pump)	OPO-Idler	Fundamental (SRS-Stokes)
Tuning range	660 ... 1010 nm	1055 ... 2340 nm	1032 nm
Output power	> 600 mW (@ 800 nm)	> 400 mW (@ 1250 nm)	> 700 mW (> 350 mW @ 20 MHz)
Pulse width	typ. 2 ps		
Spectral bandwidth	typ. 10 cm ⁻¹		
Time bandwidth product	typ. 0.6		
Repetition rate (unmodulated)	40 MHz		
Beam diameter at OPO exit	1.5 mm		
Beam divergency @ 800 nm & 1032 nm	0.8 mrad		
M ²	typ. < 1.2		
Polarization	linear / horizontal, 100:1		

Table 2.1 Optical specification of picoEmerald FT.

The common beam exit is selectable for OPO Signal and, or 1032 nm with overlap in space and time.

2.2.2 Mechanical Parameters

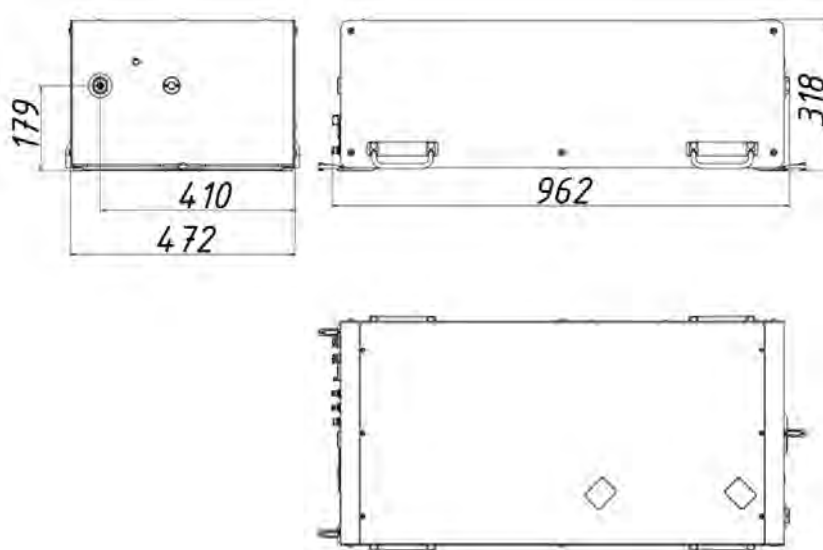


Figure 2.2 Dimensions of the optical unit (front, side and top view).

Item	Dimensions (L x W x H)	Weight
Optical unit	962 x 472 x 318 mm ³	105 kg
Power supply	19" 4 U	11 kg
Chiller	330 x 197 x 279 mm ³	10 kg

Table 2.2 Dimensions and weight of the picoEmerald FT optical unit and ancillary equipment

2.2.3 Mechanical Controls

To correct the beam overlap of the OPO signal with respect to the 1032 nm output, access openings are installed in the top lid to allow manual alignment of two mechanical mirrors mount inside the device with a 2.5 mm hex key (See figure 2.3).

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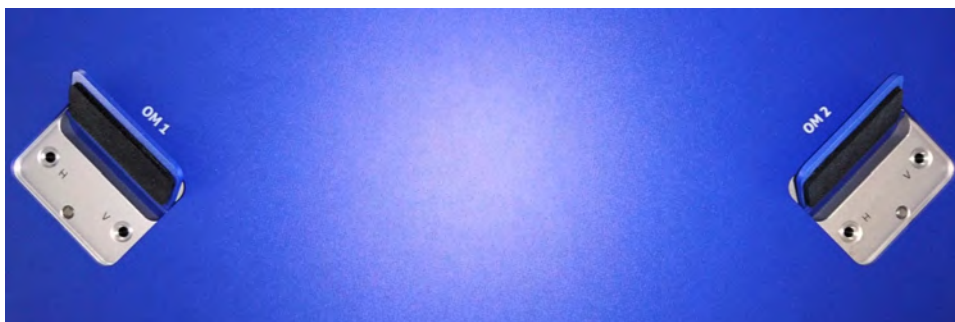


Figure 2.3 2.5 mm hex key access of top adjustment to overlap the OPO Signal (or SRS Pump) beam with 1032 nm respective SRS Stokes beam.

2.2.4 Electrical Parameters and Connectors

picoEmerald FT Power Adapter	Input: 100-240 VAC, 50/60 Hz, 1.4-0.7 A Output: 24 VDC, 5.0 A, 120 W Max.
Laser control / Power supply	100-240 VAC, 50/60 Hz, 200 W
Liquid Chiller	115/230 VAC, 60/50 Hz, 5.0/2.5 A

Table 2.3 Electrical connection data of the picoEmerald FT and ancillary equipment

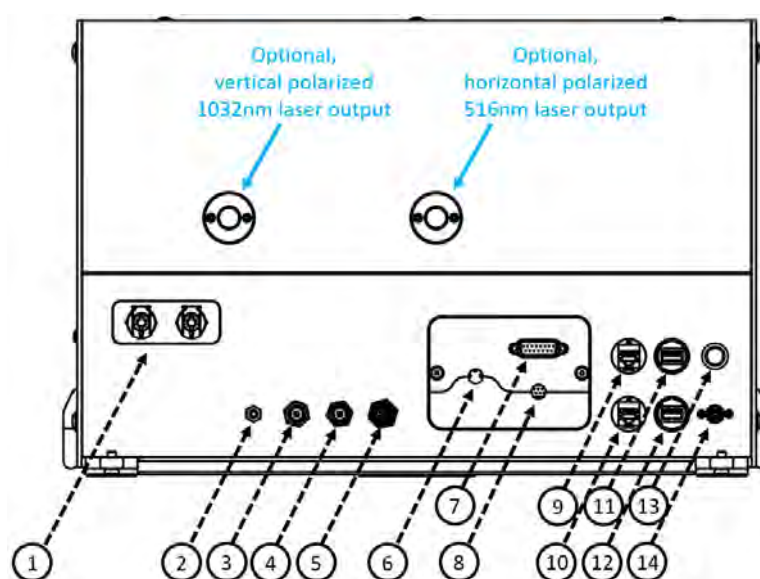


Figure 2.4 Back side of the picoEmerald FT optical unit: Electrical connections, liquid cooling and auxiliary laser outputs.

The respective connections of figure 2.4 are described in the following:

-
1. Liquid cooling connectors:
Use the delivered tubes to connect the water chiller with the picoEmerald FT optical unit. Consider the flow direction: The chiller Out has to be connected to the Cooling In plug. The tube with the filter cartridge has to be used in direction to the chiller In connector. (Please, take notice on the flow direction through the filter as marked by an arrow.)
 2. Grounding:
Use the delivered grounding cable to connect the picoEmerald FT to the electrically grounded laser table or directly on ground of the power grid.
 3. EOM Sync Out:
In case an Electro-Optical Modulator is installed a 20 MHz synchronization signal is delivered for lock-in detection.
 4. Pump Sync Out:
In case an Electro-Optical Modulator is installed the cleaned 40 MHz synchronization signal from the pulse train of the integrated 1032 nm laser is available. Otherwise the 40 MHz seed signal from areopulse is connected.
 5. Bus CAN:
CAN-bus access to add external electronic APE-modules or devices. It is important to plug the delivered CAN-bus jumper for proper operation.
 6. Power supply cable of the integrated NKT 1032 nm laser:
Connect this cable to the power supply female connector at the back side of the NKT power supply box.
 7. Seeder:
Use the delivered Seeder cable with sub-D plugs to connect the picoEmerald FT optical unit to the Seed female plug of the NKT power supply box.
 8. External Bus:
Connect this cable to the External Bus sub-D female plug of the NKT power supply box.
 9. Ethernet 1:
Use this port for direct Ethernet connection to a controller computer to run the picoEmerald FT GUI software. (The user-configured TCP/IPv4 IP address of the control computer must be set to 192.168.0.100.)
 10. Ethernet 2:
Use this port to connect the picoEmerald FT to a local network. Internet access is necessary to allow remote service.
 11. RS232 interface for remote control.
 12. Laser USB:
Use the delivered USB cable to connect the picoEmerald FT to the USB of the NKT power supply box.
 13. On/Off:
Button to start the picoEmerald FT after power connection. The status of the device is indicated by the light ring around the button:
 - (a) Status button is not lit: There is no power supplied to the optical unit.
 - (b) Status button light is red: Power supply to the optical unit is working but the picoEmerald FT is not running. In this state a running GUI software cannot connect.

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(c) Status button light is white: The picoEmerald FT is running and GUI connection is possible.

14. 24 V/5 A Power Connector:

Use the delivered AC/DC converter to supply power. Take care of the orientation of the power plug (flat side should be up).

2.2.5 Environmental Requirements

The picoEmerald FT is intended for operation in indoor, dry and dust reduced rooms. It has to be firmly installed on an optical table or on a similar solid, vibration-free board. During storage, transport, for the installation and during operation, the ambient conditions must be observed. Ensure reasonable transport conditions, free of major shocks, jolt or fall; protect against frost. Use original packing material for relocation. Before unpacking the laser wait for at least 12 hours to allow for acclimatization of all components.

Ambient temperature during transportation and storage if system is switched off (mains switch and chiller)	+ 5 ... + 50 °C
Relative humidity during transportation and storage if system is switched off (mains switch and chiller)	10 % ... 80 % (no condensation)
Ambient temperature during operation or stand by (i.e. mains switch on and chiller running)	+ 15 ... + 25 °C
Relative humidity during operation or stand by (i.e. mains switch on and chiller running)	< 60 %



If the picoEmerald FT has been stored above + 30 °C or below + 10 °C and / or above a humidity of 60 %, please establish environmental operation conditions for at least 12 hours before starting the chiller and the picoEmerald FT.
If the system has been stored for more than two weeks in the above conditions, please allow at least 24 hours in environmental operations conditions before starting the chiller and the picoEmerald FT.

Transportation at lower temperatures should be avoided. The whole cooling system needs to be completely drained and blown dry.

3 Installation

3.1 Inspection of Delivery

On the receipt of the picoEmerald FT:

1. Inspect the two packing crates for signs of rough handling or damage directly at arrival. Check the Shock Watch and Tip Tell labels on the crates:

- The Tip Tell label indicates if the angle of inclination was too high during transportation. In this case the upper white area turns (partially) blue according to the inclination angle.



inclination angle was ok



inclination angle was too high

- The Shock Watch label indicates if the succession of the shipping crate was too high during transportation. In this case the white area in the middle turns red.



succession was ok



succession was too high

2. If there are any irregularities:

- **Do not refuse to accept the package!**
- Take photographs of the condition of the package, the labels and the inside of the box, if necessary.
- List all defects on the shipping documents and let the delivery company countersign.
- Inform APE immediately.

3. If there are no irregularities, please proceed with unpacking:

- The picoEmerald FT's weight is 105 kg - four people should lift it.
- Use safe lifting practices and refer to local laws and regulations.
- Before unpacking the laser, wait at least six hours to allow for acclimatization of all components.

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The installation of the picoEmerald FT may only be carried out by an authorized field service engineer. Opening the picoEmerald FT by the user is not necessary and will void the warranty!

- The laser control unit with the chiller unit should be placed in a convenient location, away from heat-producing sources. Ensure that the intake and exhaust of the chiller are not blocked.
- picoEmerald FT and laser control unit have to be placed in a way that the on / off switches are easily accessible and its use is not restrained!
- Retain the packaging for future use.

3.2 Unpacking



Before unpacking the device wait for at least six hours (the actual time depends on the temperature difference between device and room temperature) to allow for acclimatization of all components to avoid condensation outside and inside of picoEmerald FT.

1. Remove the cover clamps of the wooden main box by using e.g. a big screwdriver as lever and open the cover plate.
2. Remove the first layer of the foam material and compare with figure 3.1. Keep the packing list and test report.



Figure 3.1 Main box for the optical unit and chiller.

3. Take the chiller unit and liquid coolant bottles (if included) out of the crate.
4. Release the rest of the box clamps to remove all side covers of the box and the rest of the foam material.

5. At each of the long sides of the picoEmerald FT there are two handles for lifting and carrying the optical unit. Use a crane or four people to lift the picoEmerald FT on the optical table and consider the orientation of the laser output. Take care to move the device smoothly and not to drop it.
6. Open the wooden supplement box of the laser controller by unscrewing the top lid.
7. Take out the two hoses.
8. Open the inner cardboard box.
9. Remove the first layer of the foam material and compare with figure 3.2 (left).



Figure 3.2 (Left) Supplement box for laser controller and accessory. (Right) Accessories of the picoEmerald FT.

10. Pick the additional components and small packages and compare with figure 3.2 (right) and the packing list.
11. Remove the next layer of the foam material and lift off the laser controller electronics.
12. Check with the components and compare with the scope of delivery (see next chapter 3.3)
13. Keep the packaging material and wooden boxes for later possible transportation.

3.3 Scope of Delivery

The standard delivery contains the picoEmerald FT, the laser control unit, the chiller, and accessory listed below:

- Power cords (3x)
- Power distributor (1x)
- Power supply for the optical unit (1x: AC/DC converter)
- USB cable type A-B (1x)

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- Ethernet cable (1x)
- RS232 null modem cable
- 15 pin sub-D cable (1x: Labelled as "Seed")
- Chiller hoses (2x, one with filter holder)
- Plastic suitcase with additional accessory, that is:
 - Filter for liquid cooling tubes (2x supplementary)
 - Open end hoses for bleed (2x)
 - 2.5 mm hex key (1x)
 - Alignment apertures for service
 - Foot clamps (3x)
 - Screws
 - Laser keys (2x)
 - Interlock plug for laser (1x)
 - CAN bus jumper plug (1x)
 - Grounding cable with plug (1x)
 - Beam shield
 - IR sensor card
 - USB flash drive with software

All these as well as supplemental components are listed in detail in an extra document received in addition to this manual. Please check if this package's content aligns with this delivery list.

3.4 Installation of the Optical Unit



Only connect all units with each other as described! Do not switch on the power of any unit!

Place the picoEmerald FT optical unit on the desired place of the optical table and fix it with the delivered clamps at each of the three feet (one in the middle of the front side and one at each side in the back). Put the laser controller unit (see figure 3.3) close by, above or below the optical table. Consider the cable length between the laser controller and the optical unit.



Figure 3.3 Front and back side of the laser controller unit.

Use table 3.1 as a guide to connect the chiller and electronics. A more detailed step by step description of the set-up is given below.

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Connectors at picoEmerald FT	Connectors at laser power supply	Others
Refer to figure 2.4	Refer to figure 3.3	
1: Liquid cooling connectors IN/OUT		Chiller OUT/IN (see figure 3.4)
2: Grounding		Screw grounding cable on the optical table.
3: EOM Sync Out		Lock-In detection (not delivered)
5: Bus CAN		Bus CAN jumper plug
6: NKT laser power supply cable	7: Power supply	
7: NKT laser seed connector	6: Seed	
8: NKT laser "External Bus" cable	3: External Bus	
9: Ethernet 1		Computer / Laptop (with static TCP/IPv4 address set to 192.168.0.100)
10: Ethernet 2		Connect to local network and internet.
11: Serial interface		
12: Laser USB	5: USB	
14: 24V/4A		Connect to power (AC/DC converter).

Table 3.1 List of associated connectors that have to be connected with the delivered cables, tubes or plugs.

- First, take the liquid cooling tubes and connect the optical unit (connections 1 in figure 2.4) with the chiller. Take care to install the provided filter cartridge inside the filter holder of the respective tube. Make sure that the following sockets are connected:
 - Connect chiller "Out" with optical unit "Cooling In". Use the tube without the filter cartridge.
 - Connect the optical unit "Cooling Out" with the chiller "In". Use the tube with the filter cartridge close to the chiller connector (see figure 3.4).
- Plug the power cable to the chiller and fill the chiller with the specified cooling liquid (Innovatek protect). Use gloves!
- Switch the power button of the chiller and continue to fill the chiller with liquid as it fills the cooling cycle of the optical unit. In case it intermediately turns off due to lack of liquid, continue filling and power cycle the chiller once the storage tank is full.
- Check that there is no water leakage between the chiller and the optical unit. Also, take a look at the connections of the filter cartridge to check for leakage.

5. The standard set temperature for the chiller is 23 °C (refer to the test report in case there's a discrepancy). Wait for the system to be at equilibrium at this temperature.
6. Continue with the following connections:
7. Plug the yellow-green grounding cable to connector 2 of figure 2.4 and screw it on the optical table. The optical table needs to be grounded as well.
8. Plug and screw the delivered CAN bus jumper to connector 5 of figure 2.4.
9. Take the power supply cable at position 6 of figure 2.4 and plug it to connector 7 of figure 3.3.
10. Take the External Bus connector cable at position 8 of figure 2.4 and plug it to connector 3 of figure 3.3.
11. Use the separately delivered 15 pin sub-D cable to link the Seed connector of the optical unit (connector 7 of figure 2.4) with the respective connector 6 of figure 3.3.
12. Use the delivered USB cable to connect the optical unit (Laser USB, connector 12 of figure 2.4) with the laser controller (USB, connector 5 of figure 3.3).
13. Connect the delivered AC/DC converter to the power grid and plug it to the 24V Power Connector of the optical unit (connector 14 of figure 2.4). Take care of the orientation of the plug (Consider the pictogram: flat side should be up!).
14. Once the picoEmerald FT is supplied with power the On/Off button (No. 13 of figure 2.4) is lit red.
15. Plug the interlock connector (or the delivered jumper) to connector 4 of figure 3.3.
16. Plug power cable to connector 2 of figure 3.3 and switch on the laser controller. In this state the laser warning lamp at the front of the picoEmerald FT is lit.
17. Put the delivered laser key into the lock 1 of figure 3.3 and turn it to "Armed" position.
18. Connect a controller computer with installed picoEmerald FT software via a local network to the picoEmerald FT. For this, use "Ethernet 2", i.e. connector 10 of figure 2.4 and an Ethernet cable. This connection is necessary for remote service via the internet. Alternatively a direct Ethernet connection to a controller computer is possible. Use the Ethernet cable to directly connect the computer to the port "Ethernet 1" (connector 9 of figure 2.4). Take care to adapt the TCP/IPv4 connection settings as described in section 4.
19. Push the On/Off button (No. 13 of figure 2.4). Its color should turn from red to white once the device is starting.
20. Install and start the delivered user software on a computer and connect it to the Ethernet port of the picoEmerald FT as described in chapter 4 "Software description". Follow the software description.

3.4.1 On/Off Button LED

The LED ring of the On/Off button indicates the following three states:

1. If the LED is not lit, the optical unit is not supplied with power.
2. If the LED light is red, the picoEmerald FT is supplied with power but the optical unit is not started.
3. If the LED light is white, the picoEmerald FT is started and ready for use. It could take several ten seconds directly after start until device connection of the GUI software is possible.

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3.5 Chiller Liquid Fill

Before starting laser operation, the chiller has to be filled with coolant. Please fill the chiller with "Innovatec Protect IP" which is provided together with the system. Connect the chiller and the picoEmerald FT with the supplied hoses. Keep the water inlet of the chiller open and switch on the chiller, while watching the water level inside the chiller (through the open water inlet). When the water level decreases, switch off the chiller and refill water to original level. Repeat this sequence, until the water level remains constant. The cooling liquid must be exchanged every six months.

The chiller set temperature should be 23 °C. Please verify, check test report and adjust if necessary.



Figure 3.4 View of the chiller with tubes, integrated cartridge filter, and water inlet (white cap). The tube with the filter should be connected to the "IN" plug of the chiller.

Please fix the cartridge filter with the supplied cable tie on the chiller handle as shown in Fig. 3.4. The cartridge must be cleaned or replaced after six months or appearance of visible impurities. If the cartridge is exchanged the cooling liquid needs to be replaced too!

A description on how to exchange the cooling liquid and the filter can be found in section 6.1.

4 Software

The installer of the user software is delivered on a USB flash drive and can be used on any computer with a Windows system. An Ethernet connection of the picoEmerald FT is necessary. There are two possibilities to establish the access:

1. For a direct connection between the computer and the picoEmerald FT the Ethernet 1 port has to be used. This port is set to a fix IP address of **192.168.0.200**. The Ethernet properties of the TCP/IPv4 connection of the computer have to be set to a user configured alternate configuration with an IP address of e.g. **192.168.0.100** and a Subnet mask of **255.255.255.0**. Please refer to chapter 6.3 for a more detailed description on how to configure your Windows 11 controller computer.
2. For software control via a DHCP managed local network Ethernet 2 port has to be used. To allow remote service, internet access is necessary.

The software setup routine generates two desktop shortcuts with respect to the users application. For CARS or SRS microscopy applications the picoEmerald FT GUI (CARS Mode) should be used. All other applications can work with the picoEmerald FT GUI (OPO Mode). The main difference of both GUIs is the labeling of controls, buttons and indicators. In the following pictures of the CARS mode are mainly shown.

Before working with the software the system should be set up and connected as described in section 3.4. To start the software double click on the picoEmerald FT GUI desktop icon.

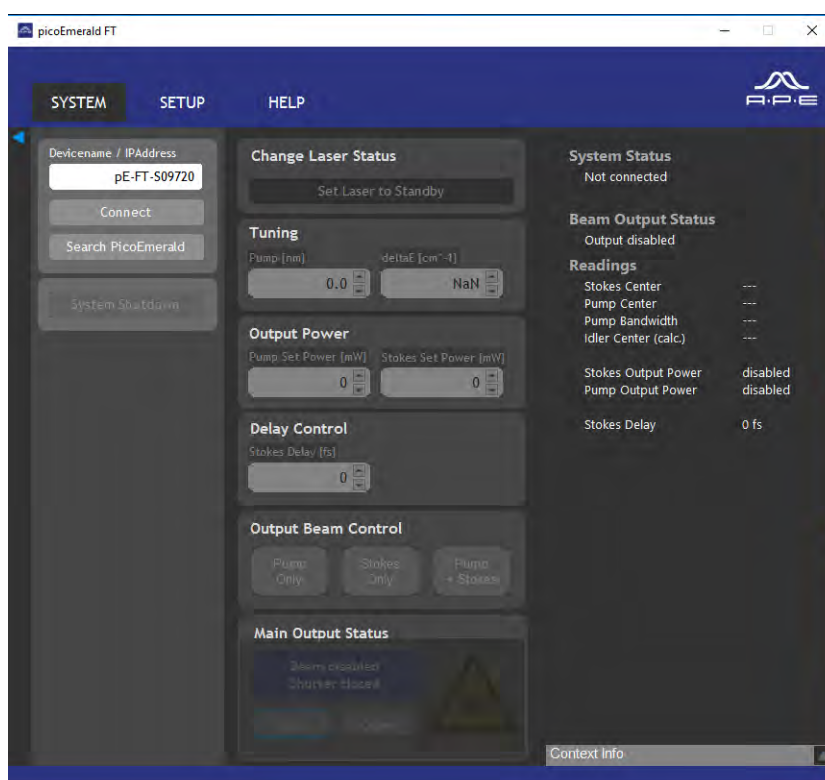


Figure 4.1 picoEmerald FT GUI (SRS and CARS MODE) after start.

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Open the SYSTEM menu of the GUI and enter the specific device name as depicted in figure 4.1 for the case of the serial number S09720. The serial number of the connected device should be used instead. Press the Connect button to continue. In case of a daily starting routine the picoEmerald will already show the system status Standby (Figure 4.2). Simply press Start Laser button, wait for a warm up time periode of around 10 minutes and start working.

4.1 Overview of picoEmerald FT's Control Software

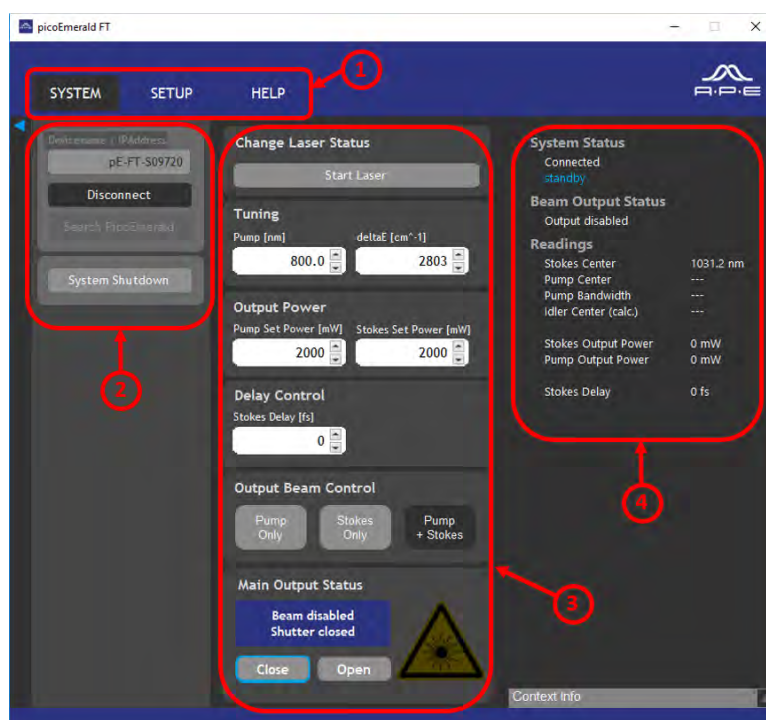


Figure 4.2 CARS Mode of the picoEmerald FT GUI after proper connection.

The user interface of the picoEmerald FT control software (Fig. 4.2) is divided in four sections:

1. Menu bar (top left):
There are three menus that can be chosen - "SYSTEM", "SETUP" and "HELP".
2. Menu specific controls (left side).
This input column can be hidden by pressing the blue triangle in the upper left.
3. Main controls (middle):
The main controls are always visible.
4. Status information column (right side):
The basic status information is always visible. Additional information can be added as described in section 4.6.

The menu specific controls are described in the following.

4.2 "SYSTEM" Menu

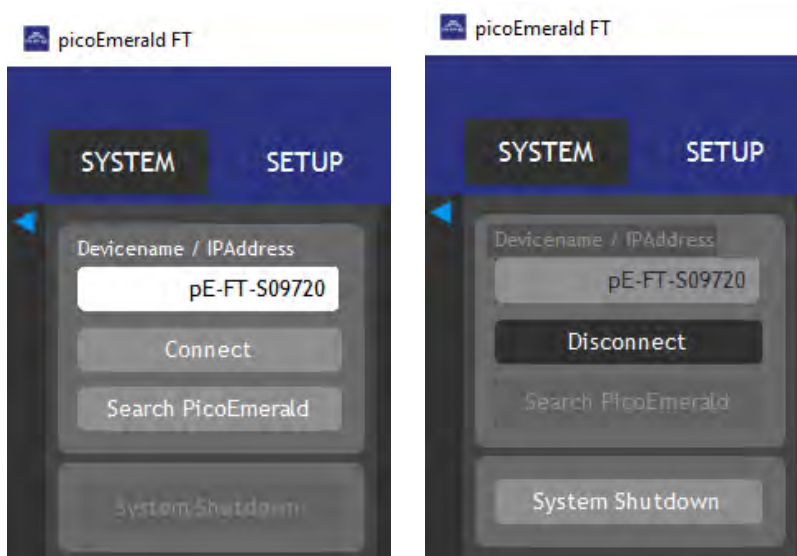


Figure 4.3 SYSTEM menu for the disconnected (left picture) and connected case (right picture).

"Devicename / IP-Address" input field

Enter the device name as shown in figure 4.3 with the specific serial number of the connected picoEmerald FT and press the Connect button (type: "pE-FT-SXXXXX" and replace the XXXXX by the serial of the device). Alternatively, the IP address of the picoEmerald FT's integrated controller computer can be used (e.g. 192.168.0.200 in case of a direct Ethernet connection to the Ethernet 1 port).

"Connect"

If the IP address or the serial number of the picoEmerald FT is entered, the button "Connect" has to be pressed. The label will change to "Disconnect" and the button will be highlighted in black. For disconnecting, press this button again (see figure 4.3). In case of a successful connection the System Status changes to "Connected" (Refer to status information column at the right side).

"Search PicoEmerald" button

If the IP address or the serial number is unknown, there is the possibility to use the search function for a picoEmerald FT that is plugged to the network. Press the "Search p'PicoEmerald" button to open a further pop-up window (see figure 4.4). This window shows a list of connected devices in which the preferred picoEmerald FT can be selected. By pressing the "Connect" button, the IP address will be entered automatically.

"System Shutdown" button

This button can only be pressed when the picoEmerald FT is connected (refer to figure 4.1 and 4.2). It has to be used in cases when the picoEmerald FT needs to be powered off and unplugged completely, e.g. because moving or transportation is necessary. A system shut down will slowly ramp down the SHG crystal temperature which takes about 20 minutes, subsequently stops the control software and turns off

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the embedded controller computer. The shut down is completed when the light ring of the picoEmerald FT's power button turns from white to red. Wait for this state, then switch off the separate aeropulse laser controller before unplugging the device.

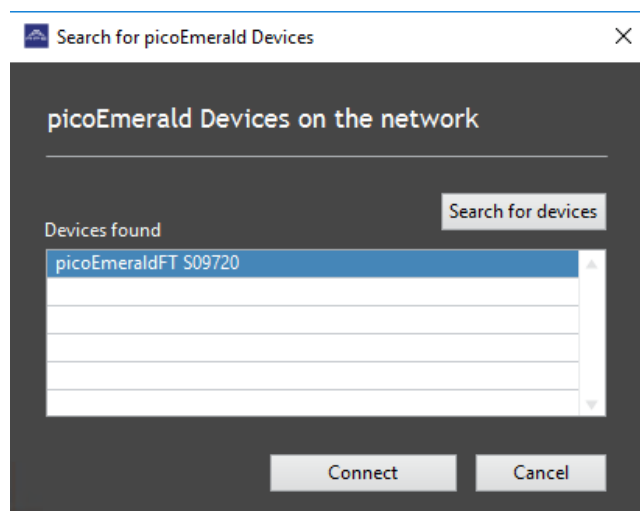


Figure 4.4 Pop-up window with a list of all picoEmerald FTs connected to the local network.

4.3 Main Control Area

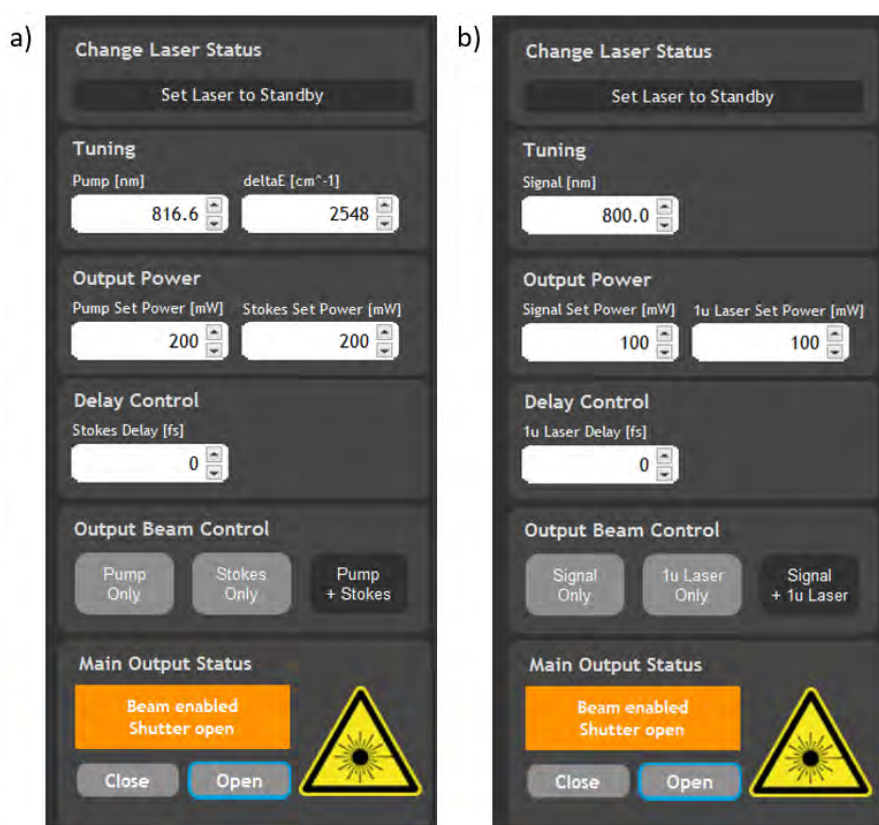


Figure 4.5 Main software controls in the middle section of the GUI [a): CARS and SRS Mode, b) OPO Mode].

"Start Laser" / "Set Laser to Standby" button

The topmost button toggles between laser start and laser standby. After successful connection of the GUI press "Start Laser" to start working. Two different sequences are running depending on the initial state of the picoEmerald FT:

1. First start after complete power off (Cold start):
The "Start Laser" action initiates a temperature ramping of the SHG crystal that is displayed in the "System Status" at the right side as "ramping up". This temperature ramping takes about 20 minutes. As soon as the crystal temperature has reached the set point the system status switches to "standby". Continue with the next step.
2. Start from system status "Connected" and "standby" (Warm start / daily routine):
Press "Start Laser" to switch on the picoEmerald FT's pump laser. Normally, this only takes some seconds and the system status changes to show "Warm Up". After a warm up periode of about 10 minutes in which power and pointing drifts are possible the system status displays "OK" and the picoEmerald FT is ready to be used.

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In case the "Start Laser" button is pressed it's function switches to "Set Laser to Standby". Press this button at the end of use (e.g. at the end of the day) to set the system to standby. The current of the integrated laser is switched off while the rest of the system and the control software keeps running.

"Tuning" controls

Enter the desired wavelength into the "Pump [nm]" (CARS Mode) or "Signal [nm]" (OPO Mode) control and press enter. The status changes from Tuning to OK as soon as the wavelength is reached. In CARS Mode of the GUI there is the alternative to enter directly the wavenumber of the molecular vibration of interest (see figure 4.5 a). In case of a picoEmerald FT with Idler option the OPO Mode (see figure 4.5 b) offers an additional control to enter a desired Idler wavelength (not shown).

"Output Power" controls

The picoEmerald FT offers independent power attenuation of the OPO signal beam (i.e. Pump beam for CARS and SRS) and the 1μ laser beam (i.e. Stokes beam). Enter the desired power values to enable automatic power adjustment after each wavelength tuning step. A set value larger than the possible maximum power value opens the attenuators completely and switches off the power adjustment procedure. The deviation of the power attenuation is typically around 1 % to 2 % but raises for low power set values (e.g. below 50 mW).

"Delay Control"

The picoEmerald FT is equipped with a software controlled delay stage for the 1μ laser pulses (i.e. Stokes pulses). The internal delay is already compensated by the software based on a factory calibrated correction function. Each time the picoEmerald FT is tuned to a new wavelength, the delay of the Stokes beam is adjusted automatically for achieving a temporal overlap between Pump and Stokes pulse at the output aperture. In addition, the effects of switchable optical components, for example filters and wave plates, are compensated as well. Enter a delay offset in femtoseconds to compensate for the effects of external optics causing dispersion. The maximum shift in positive and negative direction is about 30 ps. For positive values, the Stokes beam is delayed whereas the Pump beam is delayed for negative values.

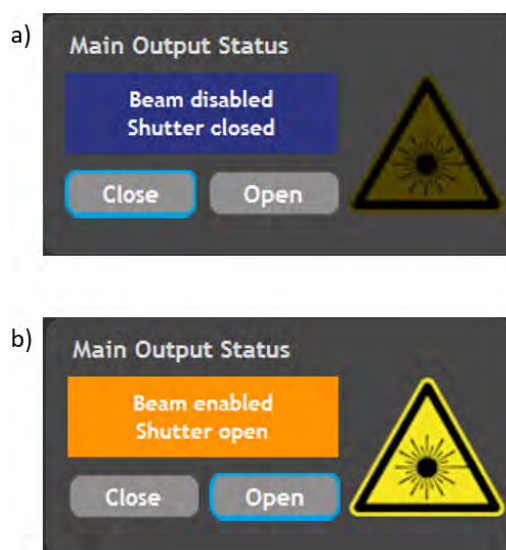


Figure 4.6 Two states of the main output shutter: a) Shutter closed, b) Shutter open.

"Output Beam Control"

The picoEmerald FT is equipped with internal beam shutters in a way that the user can select single beams or the combination of both. There are three possibilities for the main output port of the picoEmerald FT: Either the Pump beam or the Stokes beam can be emitted, or both simultaneously.

"Main Output Status"

These are the Open and Close main buttons of the picoEmerald FT. Take care, they only control the Signal and 1 μ laser (Pump and Stokes) common main exit. All other optional laser apertures are equipped with manual shutters and might therefore remain open! The current state of the main output is displayed as a text above the Close and Open buttons (See figure 4.6) . In addition for an open main exit the yellow laser warning sign is lit and the Beam Output Status at the right of the GUI shows "Output active".

4.4 "Status and Readings"

A permanent information section is displayed at the right hand side of the GUI. It displays status information and important readings of the internal sensors of the picoEmerald FT, from top to bottom:

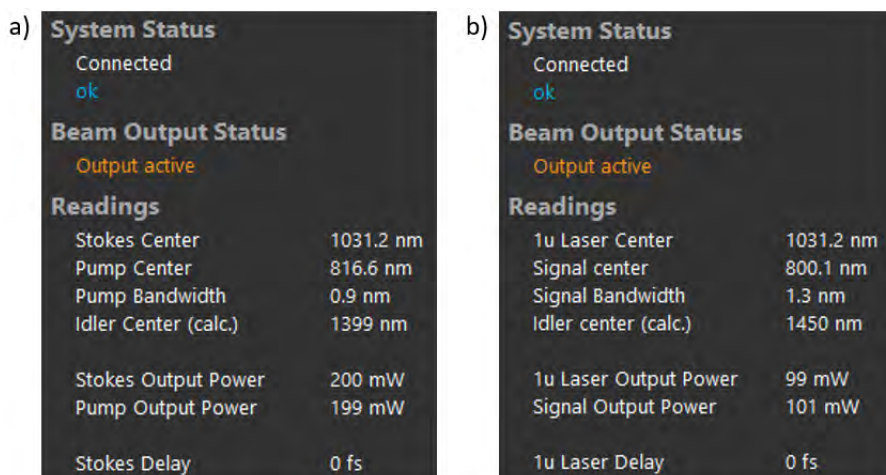


Figure 4.7 Status display and sensor readings information: a) GUI CARS and SRS Mode, b) GUI OPO Mode.

- System Status
- Beam Output Status
- Readings
 - Stokes / 1 μ Laser Center Wavelength in nm
 - Pump / Signal Center Wavelength in nm
 - Pump / Signal (FWHM) Bandwidth in nm
 - Calculated Idler Center Wavelength in nm
 - Stokes / 1 μ Laser Output Power in mW
 - Pump / Signal Output Power in mW
 - Stokes / 1 μ Laser Delay in fs

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More information about internal parameters of the picoEmerald FT is shown if "Display additional infos" is checked in the HELP menu (See section 4.6).

The following "System Status" messages are implemented (SW version 1.0.0.143):

loading	Loading procedure during software start.
no laser	There is no communication to the NKT aeropulse controller.
initializing	Actuator initialization during software start.
ready for shutdown	The SHG-crystal temperature ramping down is completed and the system is in a state for power off.
ramping up	The SHG-crystal temperature is currently ramping up to the set value.
ramping down	The SHG-crystal temperature is currently ramping down to room temperature.
standby	The NKT aeropulse laser is switched off while the system is on hold.
warming up	The NKT aeropulse laser is switched on but still in a warm up periode of approximately 15 minutes. Wait until warm up is completed before using the picoEmerald FT
optimizing shg	After warm up an optimization routine for the SHG crystal temperature is running. Wait until it is completed.
optimizing pwr	After warm up an optimization routine of the laser current is running. Wait until it is completed.
tuning	The picoEmerald FT is in a tuning routine to the set wavelength. This procedure normally takes below 3 seconds.
psd init	The routine for monitoring the spatial overlap between Pump and Stokes beams is initializing to measure the reference positions.
power adjust	The picoEmerald FT is in a tuning routine to the set wavelength and adjusts the output power.
delay adjust	The picoEmerald FT is in a tuning routine to the set wavelength and adjusts the Stokes delay.
ok	The tuning routine to the set wavelength is finished.

Table 4.1 System Status messages and explanations.

4.5 "SETUP" Menu

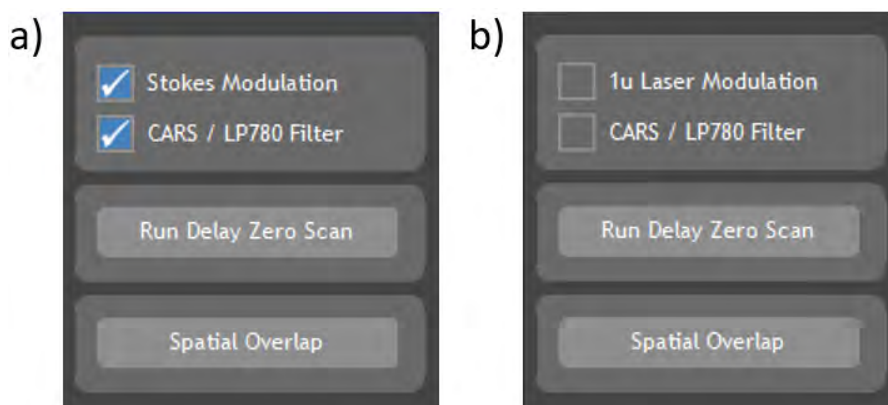


Figure 4.8 "SETUP" Menu of the picoEmerald FT's GUI (left: CARS Mode, right: OPO Mode)

"Stokes / 1 μ Laser Modulation" Checkbox

Optionally, an amplitude modulation with 20 MHz through an electro-optical modulator (EOM) is available for use with Stimulated Raman Scattering (SRS) or heterodyne CARS microscopy to suppress the non-resonant background. Synchronization signals of the laser repetition rate (40 MHz) and the modulation frequency are available at the respective outputs at the rear panel of the picoEmerald FT.

"CARS / LP780 Filter" Checkbox

To suppress distortions from residual co-propagating signals (i.e. Idler-SHG may appear below 780 nm) an additional filter can be used. In case this option is checked the software moves the filter in the beam path of the Pump (or OPO-Signal) depending on the selected wavelength. For selected wavelengths larger than 780 nm the filter moves in the beam, below 780 nm the filter is removed.

"Run Delay Zero Scan"

Press this button to run a Stokes delay scan. The internal sensor detects the zero delay position between Pump and Stokes and the delay is set to this value of zero delay.

"Spatial Overlap"

Press the "Spatial Overlap" button to open a separate pop-up window that depicts the measured beam positions of Pump and Stokes (respective OPO-Signal and 1 μ Laser) taken at two different distances within the picoEmerald FT (see figure 4.9). The beam positions are measured with two Position-Sensitive-Diodes (PSDs), one close sensor (PSD1) and a second, more distant sensor (PSD2). Directly after pressing the "Spatial Overlap" button an initialization procedure first measures the beam position of the reference Stokes beam and marks this position as dashed crosshairs. This initialization will only take a few seconds. Then the procedure switches over to continuously measure the position of the Pump beam and indicates it with an orange cross. This position can be corrected manually by aligning the top adjust mirror mounts OM1 and OM2 (see figure 2.3). Use OM1 to center the cross on the reference crosshairs of PSD1 and then do the same with OM2 on PSD2. Repeat this alignment iteratively until both beams overlap on both sensors. The spatial overlap adjustment should only be done with the Pump (OPO Signal) set to 800 nm. A difference in position of up to 20 μ m is acceptable in most cases. Normally, also a realignment when changing the wavelength is not necessary.

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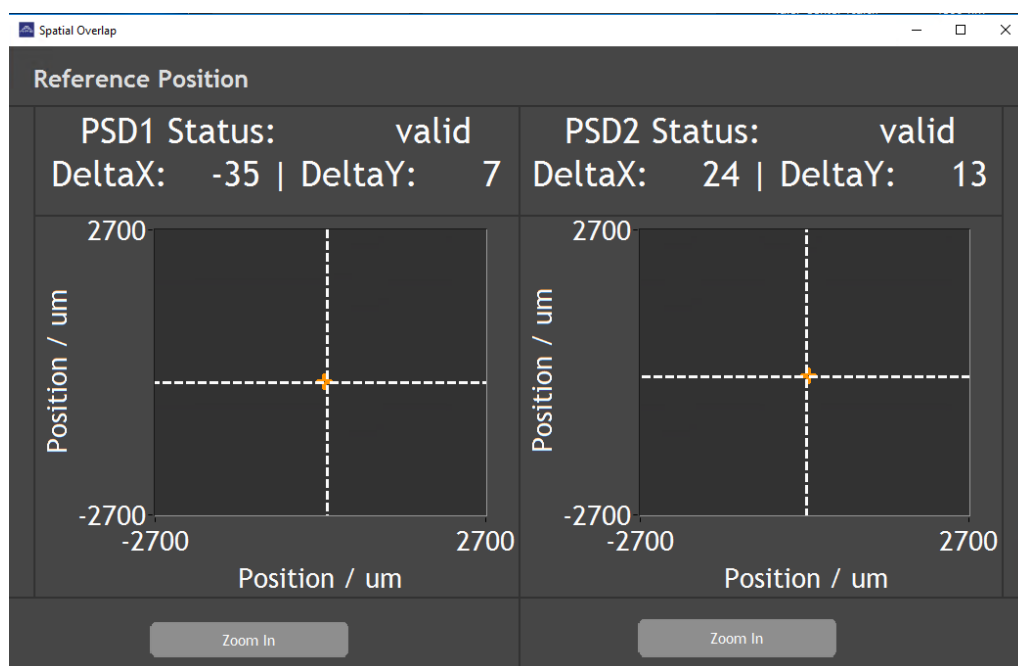


Figure 4.9 Pop-up window for spatial beam overlap alignment of Pump and Stokes.

Press the "Zoom In" buttons to magnify the beam positions for a more precise alignment.

4.6 "Help" Menu

"Display additional Infos" Checkboxes

The information section at the right hand side of the GUI can be extended with content being selectable with the check boxes (see Fig. 4.10). The 1 μ Laser Power shows the primary power of the integrated fiber laser which is commonly around 8W for standard devices. The SHG Power value shows the green pump power of the internal OPO which should normally be around 3.5W for standard devices. In addition, the temperature and humidity inside and outside of the internal cavity is measured and can be displayed. The Device Info depicts the device serial and software version numbers.

"APE Website" button

Opens the APE website on the system web browser. On the website, more information can be found on the picoEmerald FT, as well as support contact information.

"APE Calculator" button

Opens the website of the APE calculator on the system web browser. Links to the app store, to install the calculator on mobile devices can be found there too.

The APE Calculator is a software which helps to easily solve often needed (non)linear equations in optics. Some examples are the photon energy relations in Sum Frequency Generation (SFG), Difference Frequency Generation (DFG), the wavelength relations in an Optical Parametric Oscillator (OPO) as well as in coherent anti-Stokes Raman scattering (CARS) spectroscopy. Also the conversion between wavelength, wavenumber, and frequency as well as different characterizations of pulses can be calculated.



Figure 4.10 Control panel of the HELP menu.

"About" button

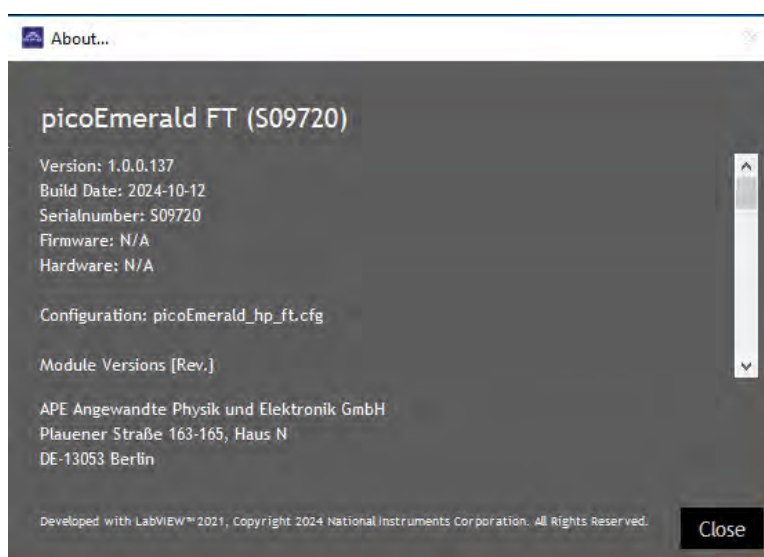


Figure 4.11 About picoEmerald FT info box.

After pressing the "About" button a popup window is shown, displaying all relevant information, e.g., serial number, firmware and software information of the picoEmerald FT device (see figure 4.11). Please refer to this information (especially the serial number) when you contact APE with questions regarding your picoEmerald FT.

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5 Daily Operation

For a long lifetime of the integrated SHG crystal it is important that it remains at working temperature. Therefore, a complete shutdown and power unplug should only be done in case of transportation of the system. The steps for such a shutdown will be described below.

5.1 Set to Standby

To turn off the laser, e.g. at the end of the day, only press the "Set Laser to Standby" software button of the GUI (see section 4.3). In case the computer has to be turned off as well, press the disconnect button and close the GUI software. If possible leave the chiller running, only turn off the chiller for a longer off period. (Account for a longer warm up period after a start with chiller switched off.) Also, leave the laser power supply switched on.

5.2 Start from Standby

When using the picoEmerald FT on a daily basis, keep the system switched on.

1. Check that the chiller is switched on and set to 23 °C. Wait for the chiller temperature to run stable at the set temperature.
2. Check that the laser power supply is switched on and the key is turned to "Armed" position.
3. Open the GUI software and go to the SYSTEM menu.
4. Enter the Device name "pE-FT-Sxxxxx" (use the serial number instead of the "xxxxx") and press the "Connect" button as described in chapter 4.
5. Press the "Start Laser" button and wait a warm up time of approximately 15 minutes.

Details of the control software are explained in chapter 4.

5.3 System Shutdown

Even in case of a longer storage without use of the device it is strongly recommended that the picoEmerald FT remains plugged to power and switched on (power button is shining white) because the internal SHG crystal is heated to avoid degradation.

A complete and proper system shutdown should only be done in case of transportation or if a power cut is planned. The whole procedure takes up to approximately 30 minutes because the internal crystal temperature is slowly ramped down before electronics switch off. For this, follow the described procedure:

1. Open the GUI software of the picoEmerald FT, connect to the device and open the System menu.
2. Press the System Shutdown button and wait approximately 30 minutes. The status depicts "ramping down".

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3. After ramping down the GUI software shows a disconnect.
4. Wait until the status light of the power button turns from white to red.
5. Switch off chiller and laser power supply.
6. Unplug the whole system.
7. Drain the whole cooling cycle (the picoEmerald FT, the tubes and empty the chiller) before packing the system for transportation.

6 Maintenance and Troubleshooting

6.1 How to exchange the chiller cooling liquid and filter cartridge

Every six months the cooling liquid must be exchanged and also the filter cartridge must be cleaned or replaced.

1. Set the picoEmerald-FT to standby, switch off the chiller and disconnect it from power.
2. Put on gloves and take a bucket.
3. Use the two separately delivered open end tubes to drain the picoEmerald-FT and the cooling tubes. For this, unplug the cooling tubes from the chiller and connect the two open end tubes to them. Take care to put one open end down into a bucket and lift the other open end higher than the picoEmerald-FT device. Wait until all the cooling liquid of the circulation is drained into the bucket.
4. Open the chiller inlet and empty the chiller tank into the bucket as well.
5. Unscrew the housing of the filter, take the filter out to replace or clean it.
6. Put in the new or cleaned filter. Take care that the sealing ring is in place and screw the filter housing back in place. Make sure not to fasten it too tight, the plastic can break!
7. Unplug the open end tubes and plug the cooling tubes back to the chiller. Take care on the input and output allocation.
8. Plug power to the chiller and continue with filling the new cooling liquid as described in section 3.5.
9. Check the water tightness of the filter cartridge and the tubes.

6.2 Reset after Interlock Trigger

In case of an interlock event the system status is set to standby and the laser is off. Once the interlock circuit is closed again a reset is done automatically by pressing the "Start Laser" software button.

6.3 How to alternate the IP configuration to control the picoEmerald-FT via Ethernet port 1

Do the following configuration steps on your Windows 11 computer to control the picoEmerald-FT via direct LAN connection to Ethernet 1 port.:

1. Open the settings menu of your computer.

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2. Open the Network and internet settings and click on Ethernet button (See figure 6.1).



Figure 6.1 Ethernet and internet settings

3. Press the Edit button of the IP assignment of the Ethernet port you want to use at your computer (See figure 6.2).

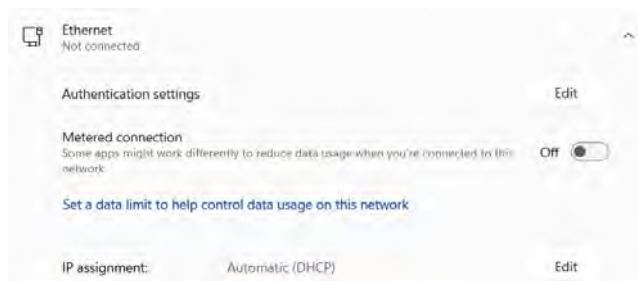


Figure 6.2 Edit IP assignment

4. Change IP settings to Manual (See figure 6.3).

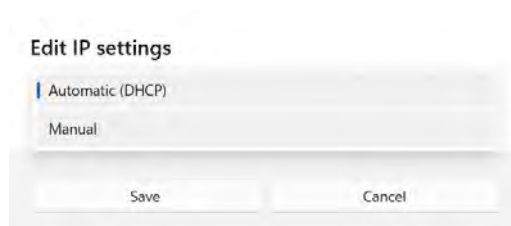


Figure 6.3 Edit IP assignment

5. Enter the fix IP address 192.168.0.100 and the Subnet mask 255.255.255.0 as shown in figure 6.4 and press Save.

Edit IP settings

Manual

IPv4

☒ On

IP address
192.168.0.100

Subnet mask
255.255.255.0

Gateway

Preferred DNS

DNS over HTTPS
Off

Alternate DNS

Save

Cancel

Figure 6.4 Edit IP assignment

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6.4 Technical Support

For technical questions or problems within Germany, please contact:

APE Angewandte Physik und Elektronik GmbH

Plauener Straße 163 - 165, Haus N

D - 13053 Berlin

+49 30 98601130 (tel)

+49 30 986011333 (fax)

service@ape-berlin.de

To contact our international distributors, please have a look to our website:

<https://www.ape-berlin.de/en/contact/>

For technical support in all other countries, please contact your local picoEmerald FT vendor.

A TCP / IP Interface

APE offers a large selection of devices capable of being used in automated measurement and experiment setups. Also the picoEmerald FT include the APE TCP interface feature allowing for remote and automated interaction. The APE TCP interface implements a standard TCP socket server enabling direct communication with the software via a defined command set. The command set of the picoEmerald FT is described in appendix B. All commands follow the SCPI99 standard, although not all features described in the standard are implemented in APE's devices.

A.1 SCPI Command Syntax

Each command has to follow the scheme shown below and must be terminated with `\r\n` (carriage return + linefeed). The device's name is optional in the commands. Each command consists of one or more keywords separated by ":". To query a value a "?" is appended and to set a value a "=" follows the command before a comma-separated list of parameters. Commands are not case sensitive.

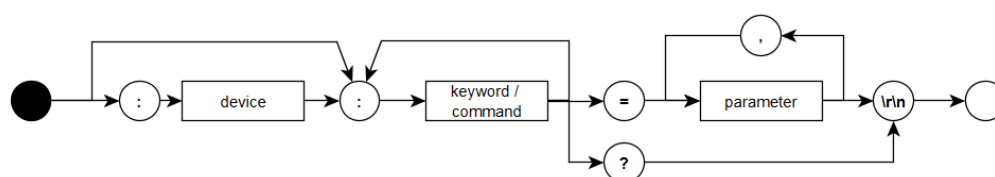


Figure A.1 SCPI command syntax scheme.

A.2 SCPI Commands - Basic Usage

Once the interface server is enabled, by sending the commands listed in the TCP SCPI Command List, it is possible to control the device or read data.

One easy way to test these commands is to use a TELNET/RAW TCP terminal. Here this is demonstrated using the application 'PuTTY' which is free, open source and available for multiple operating systems. After installing and starting the application the following window is visible:

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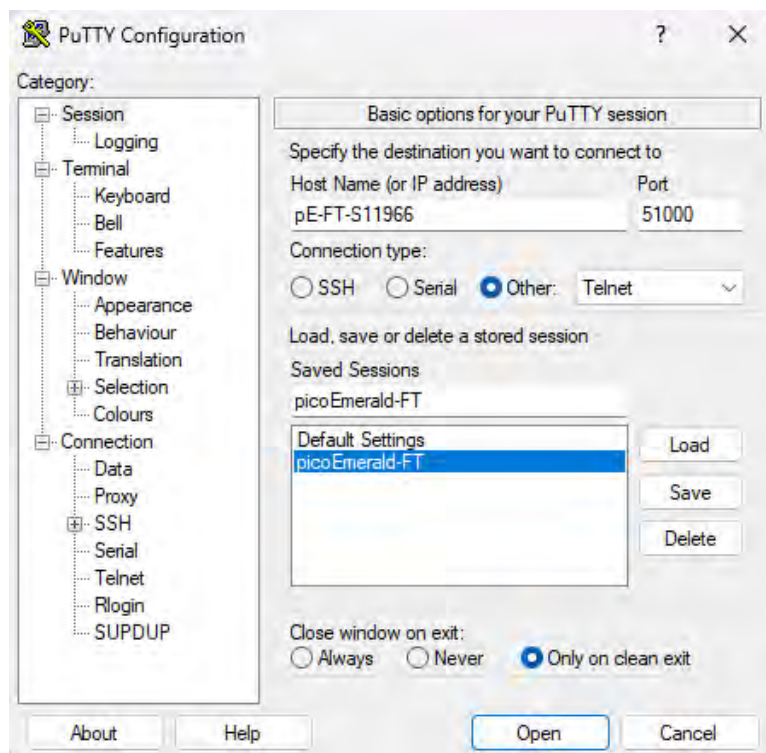


Figure A.2 PuTTY configuration window.

Enter the device's host name and the port number that is configured in the the device control application. For the picoEmerald FT the port number is set to 51000. Then select the 'TELNET' option for the connection type and click the "Open" button to establish the connection.

If the connection was established successfully a black terminal window appears. In case of an error the window also opens, but an error message appears and shows further information.

The commands can be typed into the window same as they are described in the command list. To send the command, the RETURN key has to be pressed. If the command returns a value, the output will appear below the line that contains the just entered command.

The picture below shows the *idn? command as an example. The command is written in the first line and the interface's response (device information string) in the second line.

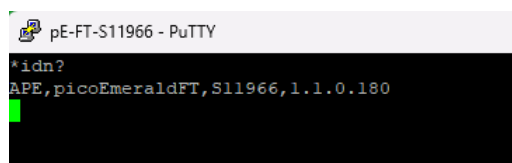


Figure A.3 PuTTY terminal window, *idn? command and response.

A.3 SCPI Block Handling

Some commands return data in binary form in a so called SCPI block. Block data is transferred using a specific header format described in the standard. The response layout looks like this:

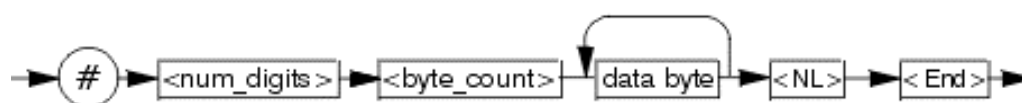


Figure A.4 SCPI data block syntax scheme.

The header starts with "#", followed by a single character with the number of digits needed to specify the total byte count and then this many characters with the total byte count. After the header, the previously given number of bytes follow suffixed with `\r\n`. The string "TEST" is encoded to an SCPI block "#14TEST\r\n" while "TESTTESTTEST" becomes "#12TESTTESTTEST\r\n". In the first example, the number of data bytes is four, so the byte count can be represented in one digit. In the second example, the number of data bytes is twelve, so the byte count needs two digits for representation.

The data bytes can hold any form of data. APE devices return spectra or autocorrelation data in binary form that must be unpacked into usable array(s). APE provides example code snippets that demonstrate this in different programming languages. The example code can be downloaded here: <https://www.ape-berlin.de/software-interface-tcpip/>.

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B TCP/IP Commands for Remote Communication and Automatization

*IDN?

Get Device Identification

<manufacturer>	string
	Manufacturer Name
<model>	string
	Modelnumber/-name
<software>	string
	Software Version
<firmware>	string
	Firmware Revision

Example:

```
*IDN?
```

*RST

Execute device reset

Example:

```
*RST
```

INITIALIZED?

Returns the internal initialization state. When the SW was just started, or the *RST Function was used, the device will reset this state to FALSE until the startup of all internal modules has been completed. While this command returns FALSE the other commands for the device will not work.

<status>	boolean
	Status of the highlevel initialization

Example:

```
initialized?
```

*STB?

Get Device Status Byte

<parameter>	integer
	SCPI Device Status Byte

Example:

```
*STB?
```

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*ESR?

Get event status register

<esr> integer
SCPI Event Status Register

Example:

*ESR?

*OPER?

Get operation status register

<oper> integer
SCPI Operation Status Register

Example:

*OPER?

LIST_MODULES?

MDLS?

Get list of all loaded modules (name, type, version)

<modules> string
List of all loaded modules, Format = [MODULE_NAME] (T:
[MODULE_TYPE], R: [FIRMWARE/SVN_REVISION])

Example:

list_modules?

INTERFACE_XML?

IXML?

Get APE Interface Description in XML Format

<xml_file> array of bytes in block data format
SCPI-Block-Data; XML-File-Content is passed as ASCII Values in
Block-Data-Segment

Example:

interface_xml?

CFG_FILE? CFG?

Dump the currently used configuration file to the tcp socket
 <cfg_file> array of bytes in block data format
 SCPI-Block-Data; Configuration-File-Content is passed as ASCII Values
 in Block-Data-Segment

Example:

```
cfg_file?
```

SERVICEMODE<status> <salt> <hash>

Set new service level. Salt returned by reading this command is required to set the servicemode

<status> integer
 <salt> string
 <hash> string

Example:

```
servicemode=[Salted Hash]
```

SERVICEMODE?

Get current service level

<status> integer
 Service level (0 - User, 1 = Servicemode)
 <salt> string
 Salt required for setting the servicemode
 <hash> string
 SHA256-Hash as Base64-String over "[Salt] + [Password for Servicemode]". Salt is returned by reading this command

Example:

```
servicemode?
```

WAVELENGTH<wavelength>

Set signal wavelength

<wavelength> double, unit: nm

Example:

```
wavelength=800.0
```

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WAVELENGTH?

Get current signal wavelength
<wavelength> double, unit: nm
wavelength in nm

Example:

SET_WAVELENGTH<set-wavelength>

Set signal wavelength
<set-wavelength> double, unit: nm

Example:

SET_WAVELENGTH?

Get set-signal wavelength
<set-wavelength> double, unit: nm
wavelength in nm

Example:

TUNING<tuning>

Set tuning hold
<tuning> double

Example:

TUNING?

Get tuning hold
<tuning> double
tuning hold state

Example:

SIGNAL_POWER<signal-power>

Set signal power
<signal-power> double, unit: mW

Example:

```
signal_power=100
```

SIGNAL_POWER?

Get current signal set power
<signal-power> double, unit: mW
power in mW

Example:

```
signal_power?
```

SET_SIGNAL_POWER<set-signal-power>

Set signal power
<set-signal-power> double, unit: mW

Example:

```
set_signal_power=100
```

SET_SIGNAL_POWER?

Get set-signal power
<set-signal-power> double, unit: mW
power in mW

Example:

```
set_signal_power?
```

SIGNAL_PWR_HOLD<tuning>

Set signal power hold
<tuning> double

Example:

```
signal_pwr_hold=0
```

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SIGNAL_PWR_HOLD?

Get signal power hold
<tuning> double
signal power hold state

Example:

```
signal_pwr_hold?
```

SIGNAL_SHUTTER<signal-shutter-position>

Set Signal Shutter
<signal-shutter-position> double

Example:

```
signal_shutter=1
```

SIGNAL_SHUTTER?

Get Signal shutter position
<signal-shutter-position> double
signal shutter position (0 = close, 1 = open)

Example:

```
signal_shutter?
```

SET_SIGNAL_SHUTTER<set-signal-shutter-position>

Set set-Signal Shutter
<set-signal-shutter-position> double

Example:

```
set_signal_shutter=1
```

SET_SIGNAL_SHUTTER?

Get set-Signal shutter position
<set-signal-shutter-position> double
set signal shutter position (0 = close, 1 = open)

Example:

```
set_signal_shutter?
```

FILTER<filter>

Set filter state
<filter> double

Example:

filter=1

FILTER?

Get filter state
<filter> double
filter state (0 = off, 1 = on)

Example:

filter?

IR_POWER<ir-power>

Set IR power
<ir-power> double, unit: mW

Example:

ir_power=100

IR_POWER?

Get current IR power
<ir-power> double, unit: mW
power in mW

Example:

ir_power?

SET_IR_POWER<set-ir-power>

Set IR power
<set-ir-power> double, unit: mW

Example:

set_ir_power=100

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SET_IR_POWER?

Get set-ir power
<set-ir-power> double, unit: mW
power in mW

Example:

IR_PWR_HOLD<tuning>

Set ir power hold
<tuning> double

Example:

IR_PWR_HOLD?

Get ir power hold
<tuning> double
ir power hold state

Example:

IR_SHUTTER<power>

Set IR Shutter
<power> double

Example:

IR_SHUTTER?

Get IR Shutter position
<power> double
ir shutter position (0 = close, 1 = open)

Example:

SET_IR_SHUTTER<set-ir-shutter-position>

Set set-IR Shutter
<set-ir-shutter-position> double

Example:

```
set_ir_shutter=1
```

SET_IR_SHUTTER?

Get set-IR Shutter position
<set-ir-shutter-position> double
set ir shutter position (0 = close, 1 = open)

Example:

```
set_ir_shutter?
```

MAIN_SHUTTER<power>

Set Main Shutter
<power> double

Example:

```
main_shutter=1
```

MAIN_SHUTTER?

Get main shutter position
<power> double
main shutter position (0 = close, 1 = open)

Example:

```
main_shutter?
```

MODULATION<modulation>

Set Modulation state
<modulation> double

Example:

```
modulation=1
```

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MODULATION?

Get Modulation state
<modulation> double
modulation state (0 = off, 1 = on)

Example:

DELAY<delay>

Set delay
<delay> double, unit: fs

Example:

DELAY?

Get delay
<delay> double, unit: fs
delay in fs

Example:

SET_DELAY<delay>

Set delay
<delay> double, unit: fs

Example:

SET_DELAY?

Get set-delay
<delay> double, unit: fs
set_delay in fs

Example:

ENABLE_DELAY<delay-state>

Set enable-delay
<delay-state> double

Example:

```
enable_delay=0
```

ENABLE_DELAY?

Get enable-delay
<delay-state> double
delay state

Example:

```
enable_delay?
```

PARAMETER?

Get Parameters
<parameters> string
parameters (state, center, fwhm, signal_poer, signal_shuter,
stokes1_power, stokes1_sgutter, stokes2_power, stokes2_shutter)

Example:

```
parameter?
```